

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 4: Single Equation Models of Multiple Time Series

Fu Ouyang

School of Economics
University of Queensland

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Multiple Time Series Models

Recommended readings

Author	Title	Chapter	Call No
Patterson	An Introduction to Applied Econometrics: A Time Series Approach	2 (except 2.6)	HB139 .P373 2000
Enders	Applied Econometric Time Series, 4e	5.1-5.3	HB139 .E55 2015
Hill et al.	Principles of Econometrics	9	HB139 .H548 2008

Autoregressive Distributed Lag

In this lecture, we explore the **autoregressive distributed lag** (ARDL) model as a way to jointly model two or more variables (e.g. y_t, x_t, w_t).

Today's lecture is an introduction to several topics coming in the following weeks:

- stochastic trends (next week)
- equilibrium error correction
- multiple equation models

The ARDL encompasses many other models.

The ARDL Model

The ARDL model uses lags of both y_t and x_t to permit richer modeling of dynamics.

The ARDL(p, l) general form:

$$\begin{aligned} y_t &= a_0 + a_1 y_{t-1} + \cdots + a_p y_{t-p} + \theta_0 x_t + \theta_1 x_{t-1} + \cdots + \theta_l x_{t-l} + \varepsilon_t, \\ &= a_0 + \sum_{i=1}^p a_i y_{t-i} + \sum_{j=0}^l \theta_j x_{t-j} + \varepsilon_t. \end{aligned}$$

Examples:

$$\text{ARDL}(2, 2) : \quad y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \theta_0 x_t + \theta_1 x_{t-1} + \theta_2 x_{t-2} + \varepsilon_t,$$

$$\text{ARDL}(4, 1) : \quad y_t = a_0 + a_1 y_{t-1} + \cdots + a_4 y_{t-4} + \theta_0 x_t + \theta_1 x_{t-1} + \varepsilon_t.$$

The ARDL with Several Variables

Using **lag operator** notation, the ARDL(p, l) can be written:

$$a(L)y_t = a_0 + \theta(L)x_t + \varepsilon_t.$$

This is useful to represent ARDLs with **several** variables—e.g., an ARDL(p, l, s) is given by:

$$a(L)y_t = a_0 + \theta(L)x_t + \gamma(L)w_t + \varepsilon_t,$$

where

$$a(L) = 1 - a_1L - \cdots - a_pL^p,$$

$$\theta(L) = \theta_0 + \theta_1L + \cdots + \theta_lL^l,$$

$$\gamma(L) = \gamma_0 + \gamma_1L + \cdots + \gamma_sL^s.$$

Dynamic Effects - Multipliers in the ARDL

ARDL allows us to construct IRFs of y_t to shocks in either x_t or y_t .

Example: ARDL(1, 1) given by $y_t = a_0 + a_1 y_{t-1} + \theta_0 x_t + \theta_1 x_{t-1} + \varepsilon_t$.

1. Immediate response or **impact multiplier**:

$$\frac{\partial y_t}{\partial x_t} = \theta_0.$$

2. The response after one period, two periods, etc.:

$$\begin{aligned}\frac{\partial y_{t+1}}{\partial x_t} &= a_1 \frac{\partial y_t}{\partial x_t} + \theta_1 = a_1 \theta_0 + \theta_1, \\ \frac{\partial y_{t+2}}{\partial x_t} &= a_1 \frac{\partial y_{t+1}}{\partial x_t} = a_1 (a_1 \theta_0 + \theta_1), \\ &\vdots\end{aligned}$$

Dynamic Effects - Multipliers in the ARDL

The effect on y_{t+h} is decreasing as long as $|a_1| < 1$ (stability condition).

The **long run multiplier** (LRM) is the sum of effects over time:

Given the stability condition is satisfied, LRM in the ARDL(1, 1) is

$$\begin{aligned}\text{LRM} &= \theta_0 + a_1\theta_0 + \theta_1 + a_1(a_1\theta_0 + \theta_1) + \cdots, \\ &= (1 + a_1 + a_1^2 + \cdots)(\theta_0 + \theta_1), \\ &= \frac{\theta_0 + \theta_1}{1 - a_1}.\end{aligned}$$

We return to the long-run effect later in the lecture.

Multipliers ARDL(1, 1) - Summary

RECAP	Time (t)	Multiplier
1) Immediate response	$t = 0$	θ_0
2) The effect after 1-period	$t + 1$	$a_1\theta_0 + \theta_1$
3) The effect after 2-periods	$t + 2$	$a_1(a_1\theta_0 + \theta_1)$
Sum 1) + 2) + 3) + $\dots$	LRM	$\frac{\theta_0 + \theta_1}{1 - a_1}$

The ARDL Family of Models

$$\text{ARDL}(1, 1): y_t = a_0 + a_1 y_{t-1} + \theta_0 x_t + \theta_1 x_{t-1} + \varepsilon_t$$

1. Static Regression:

$$y_t = a_0 + \theta_0 x_t + \varepsilon_t;$$

Restrictions: $a_1 = 0, \theta_1 = 0$.

2. First-order autoregression:

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t;$$

Restrictions: $\theta_0 = 0, \theta_1 = 0$.

3. Leading indicator equation:

$$y_t = a_0 + \theta_1 x_{t-1} + \varepsilon_t;$$

Restrictions: $a_1 = 0, \theta_0 = 0$.

4. Equation in first differences:

$$\Delta y_t = a_0 + \theta_0 \Delta x_t + \varepsilon_t;$$

Restrictions: $a_1 = 1, \theta_1 = -\theta_0$.

The ARDL family of Models

$$\text{ARDL}(1, 1): y_t = a_0 + a_1 y_{t-1} + \theta_0 x_t + \theta_1 x_{t-1} + \varepsilon_t$$

5. First-order distributed lag:

$$y_t = a_0 + \theta_0 x_t + \theta_1 x_{t-1} + \varepsilon_t;$$

Restrictions: $a_1 = 0$.

6. Partial adjustment model:

$$y_t = a_0 + a_1 y_{t-1} + \theta_0 x_t + \varepsilon_t;$$

Restrictions: $\theta_1 = 0$.

7. Dead start model:

$$y_t = a_0 + a_1 y_{t-1} + \theta_1 x_{t-1} + \varepsilon_t;$$

Restrictions: $\theta_0 = 0$.

8. Proportional response model:

$$y_t = a_0 + a_1 (y_{t-1} - x_{t-1}) + \theta_0 x_t + \varepsilon_t;$$

Restrictions: $a_1 = 1, \theta_1 = -a_1$.

9. Error correction model:

$$\Delta y_t = a_0 + \alpha (y_{t-1} - \beta x_{t-1}) + \theta_0 \Delta x_t + \varepsilon_t;$$

This is a re-arrangement of the ARDL equation.

The Error Correction Model (ECM)

Start with the ARDL(1, 1):

$$y_t = a_0 + a_1 y_{t-1} + \theta_0 x_t + \theta_1 x_{t-1} + \varepsilon_t.$$

Add and subtract y_{t-1} and $\theta_0 x_{t-1}$:

$$\begin{aligned} y_t &= a_0 + a_1 y_{t-1} + y_{t-1} - y_{t-1} \\ &\quad + \theta_0 x_t + \theta_0 x_{t-1} - \theta_0 x_{t-1} + \theta_1 x_{t-1} + \varepsilon_t, \end{aligned}$$

then re-arrange:

$$\begin{aligned} y_t - y_{t-1} &= a_0 + (a_1 - 1)y_{t-1} + (\theta_0 + \theta_1)x_{t-1} + \theta_0(x_t - x_{t-1}) + \varepsilon_t, \\ \Delta y_t &= a_0 + (a_1 - 1)y_{t-1} + (\theta_0 + \theta_1)x_{t-1} + \theta_0 \Delta x_t + \varepsilon_t, \\ \Delta y_t &= a_0 + \alpha(y_{t-1} - \beta x_{t-1}) + \theta_0 \Delta x_t + \varepsilon_t, \quad \text{this is the ECM.} \end{aligned}$$

where $\alpha = -(1 - a_1)$ and $\beta = \frac{\theta_0 + \theta_1}{1 - a_1}$.

The ECM and Long-Run Equilibrium

The ECM has a particularly important interpretation in terms of the long-run equilibrium.

Two non-stationary variables y_t and x_t are said to co-exist in long-run equilibrium if $z_t = y_t - \beta x_t$ is stationary (with $E(z_t) = 0$).

In the short-run shocks to y_{t-1} and/or x_{t-1} cause the variables to deviate from the long-run equilibrium, such that $y_{t-1} - \beta x_{t-1} \neq 0$.

Since the long-run relationship is stationary, y_t must adjust to restore equilibrium; the adjustment Δy_t is proportional to the magnitude of the deviation, i.e. $\alpha(y_{t-1} - \beta x_{t-1})$.

α determines the speed of adjustment.

The ECM and Long-Run Equilibrium

The **equilibrium solution** of the ARDL is the **mean path of (stationary) y_t when x_t is constant**.

The **equilibrium** between y_t and x_t can be found by setting them all to their long-run values and $\varepsilon_t = 0$:

$$y_t = y_{t-1} = y^* \qquad x_t = x_{t-1} = x^*.$$

Substituting into the ARDL(1, 1), we obtain:

$$y^* = a_0 + a_1 y^* + \theta_0 x^* + \theta_1 x^*,$$
$$y^* = \frac{a_0}{1 - a_1} + \frac{\theta_0 + \theta_1}{1 - a_1} x^*.$$

The **mean path response** is therefore $y^* = \mu + \beta x^*$, where β is the LRM.

Illustration

Consider the problem of estimating the relationship between quarterly capital expenditures and appropriations for manufacturing firms.

The data are from Hill *et al.* (2001), Chapter 15.

The model is ARDL(p, l):

$$y_t = a_0 + \sum_{i=1}^p a_i y_{t-i} + \sum_{j=0}^l \theta_j x_{t-j} + \varepsilon_t,$$

where

y_t : quarterly capital expenditures,

x_t : appropriations,

$T = 88$ quarters.

Output for ARDL(1, 1)

Dependent Variable: Y

Method: Least Squares

Sample (adjusted): 2 88

Included observations: 87 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-2.403615	28.95567	-0.083010	0.9340
X	0.034779	0.018967	1.833658	0.0703
X(-1)	0.156188	0.023504	6.645260	0.0000
Y(-1)	0.800800	0.017010	47.07956	0.0000
R-squared	0.997501	Mean dependent var	4331.851	
Adjusted R-squared	0.997410	S.D. dependent var	2196.238	
S.E. of regression	111.7642	Akaike info criterion	12.31555	
Sum squared resid	1036772.	Schwarz criterion	12.42892	
Log likelihood	-531.7262	Hannan-Quinn criter.	12.36120	
F-statistic	11041.90	Durbin-Watson stat	1.549251	
Prob(F-statistic)	0.000000			

Output for ARDL(1, 2)

Dependent Variable: Y

Method: Least Squares

Sample (adjusted): 3 88

Included observations: 86 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.031351	27.90879	-0.001123	0.9991
X	0.042851	0.018318	2.339324	0.0218
X(-1)	0.068367	0.036297	1.883555	0.0632
Y(-1)	0.758669	0.021178	35.82410	0.0000
X(-2)	0.118692	0.038246	3.103414	0.0026
R-squared	0.997750	Mean dependent var		4358.070
Adjusted R-squared	0.997639	S.D. dependent var		2195.381
S.E. of regression	106.6768	Akaike info criterion		12.23387
Sum squared resid	921775.5	Schwarz criterion		12.37656
Log likelihood	-521.0562	Hannan-Quinn criter.		12.29129
F-statistic	8979.669	Durbin-Watson stat		1.545836
Prob(F-statistic)	0.000000			

Output for ARDL(2, 1)

Dependent Variable: Y

Method: Least Squares

Sample (adjusted): 3 88

Included observations: 86 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	5.707282	27.67370	0.206235	0.8371
X	0.037896	0.017908	2.116172	0.0374
X(-1)	0.108816	0.026142	4.162494	0.0001
Y(-1)	1.128740	0.095923	11.76720	0.0000
Y(-2)	-0.284722	0.082128	-3.466813	0.0008
R-squared	0.997808	Mean dependent var		4358.070
Adjusted R-squared	0.997699	S.D. dependent var		2195.381
S.E. of regression	105.2988	Akaike info criterion		12.20786
Sum squared resid	898115.2	Schwarz criterion		12.35056
Log likelihood	-519.9381	Hannan-Quinn criter.		12.26529
F-statistic	9216.766	Durbin-Watson stat		2.265866
Prob(F-statistic)	0.000000			

Output for ARDL(2, 2)

Dependent Variable: Y

Method: Least Squares

Sample (adjusted): 3 88

Included observations: 86 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.520561	26.90555	0.279517	0.7806
X	0.043773	0.017575	2.490608	0.0148
X(-1)	0.049477	0.035453	1.395584	0.1667
Y(-1)	1.037766	0.100637	10.31201	0.0000
X(-2)	0.091098	0.037961	2.399744	0.0187
Y(-2)	−0.233847	0.082584	−2.831613	0.0059
R-squared	0.997955	Mean dependent var	4358.070	
Adjusted R-squared	0.997827	S.D. dependent var	2195.381	
S.E. of regression	102.3356	Akaike info criterion	12.16161	
Sum squared resid	837806.1	Schwarz criterion	12.33284	
Log likelihood	−516.9491	Hannan-Quinn criter.	12.23052	
F-statistic	7807.755	Durbin-Watson stat	2.143801	
Prob(F-statistic)	0.000000			

Illustration - Model Comparison

Model Information Criteria Computed Values:

ARDL(1,1)	AIC	12.3155
	SC	12.4289
ARDL(2,1)	AIC	12.2978
	SC	12.3505
ARDL(1,2)	AIC	12.2338
	SC	12.3765
ARDL(2,2)	AIC	12.1616*
	SC	12.3328*

ARDL Estimation

Dependent Variable: Y

Method: ARDL

Sample (adjusted): 3 88

Included observations: 86 after adjustments

Dependent lags: 2 (Fixed)

Dynamic regressors (2 lags, fixed): X

Fixed regressors: C

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
Y(-1)	1.037766	0.100637	10.31201	0.0000
Y(-2)	-0.233847	0.082584	-2.831613	0.0059
X	0.043773	0.017575	2.490608	0.0148
X(-1)	0.049477	0.035453	1.395584	0.1667
X(-2)	0.091098	0.037961	2.399744	0.0187
C	7.520561	26.90555	0.279517	0.7806
R-squared	0.997955	Mean dependent var		4358.070
Adjusted R-squared	0.997827	S.D. dependent var		2195.381
S.E. of regression	102.3356	Akaike info criterion		12.16161
Sum squared resid	837806.1	Schwarz criterion		12.33284
Log likelihood	-516.9491	Hannan-Quinn criter.		12.23052
F-statistic	7807.755	Durbin-Watson stat		2.143801
Prob(F-statistic)	0.000000			

ARDL Estimation

This feature can be used to compute the LRM.

ARDL Cointegrating And Long Run Form

Original dep. variable: Y

Selected Model: ARDL(2, 2)

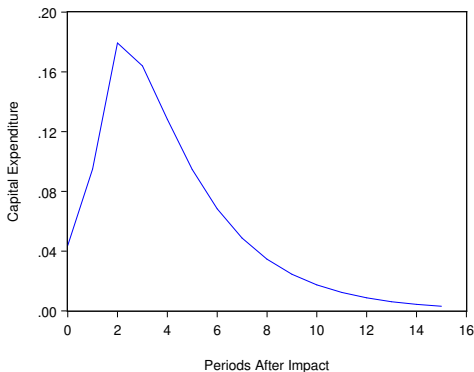
Sample: 1 88

Included observations: 86

Cointegrating Form				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(Y(-1))	0.233847	0.080740	2.896278	0.0049
D(X)	0.043773	0.017449	2.508704	0.0141
D(X(-1))	-0.091098	0.037726	-2.414720	0.0180
C	7.520561	12.094235	0.621830	0.5358
CointEq(-1)	-0.196080	0.023871	-8.214106	0.0000
Cointeq = Y - (0.9402*X)				

Long Run Coefficients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
X	0.940168	0.030294	31.034693	0.0000

Estimated Marginal Effects from ARDL(2, 2)



Total multiplier: 0.9402 $\implies$ for each dollar in appropriation money, 0.94 cents are spent on capital expenditures.